

Design and Implementation of an Intelligent Food Waste Analysis Agent

1. Introduction

Food waste is a critical global issue that impacts economic sustainability, environmental protection, and social welfare. Large quantities of edible food are discarded due to poor forecasting, inefficient supply chains, and lack of intelligent decision support. This project presents an intelligent agent designed to analyze food waste data, learn consumption patterns, and support decision-making through prediction and reasoning. The agent is developed using the Food Waste dataset from Kaggle and integrates multiple artificial intelligence techniques covered during the course.

2. Problem Definition

The problem addressed is the intelligent analysis of historical food waste data to anticipate future waste levels and identify potentially critical situations. Rather than outputting raw predictions, the agent reasons about outcomes and selects appropriate actions such as monitoring, issuing alerts, or recommending waste reduction strategies.

2.1 Problem Domain

The domain involves food waste monitoring across different food categories and time periods. The environment is uncertain due to seasonal effects, consumer behavior, and operational variability. The agent must learn from historical observations while reasoning under incomplete and noisy data.

2.2 Formal Problem Specification

State Representation: $s = (F, W, t)$, where F denotes food category, W the waste quantity, and t the time index. The initial state corresponds to the earliest dataset entry. Actions include Monitor, Recommend Reduction, and Alert High Waste. The transition model represents temporal progression combined with predicted waste values. There is no terminal goal state; performance is measured through accuracy and decision quality.

3. Dataset Description

The Food Waste dataset contains structured records representing food waste quantities over time. It includes numerical and categorical features reflecting real-world variability. Preprocessing steps include handling missing values, encoding categorical variables, and normalizing numerical features.

4. Methodology

Machine Learning Component: A supervised learning model is trained to predict future waste levels or classify waste intensity. The model captures non-linear relationships between food categories and waste trends.

Reasoning Component: A rule-based decision layer evaluates predictions using predefined thresholds and heuristics. This component selects actions that minimize potential waste and improve interpretability.

5. Evaluation

The system is evaluated using prediction accuracy, classification reliability, and the correctness of decision outcomes. Results show that combining learning-based predictions with reasoning significantly reduces false alerts and improves practical usefulness compared to a standalone predictive model.

6. Ethical Considerations

Automated food waste analysis systems may influence operational and policy decisions. Incorrect predictions could lead to poor resource allocation. To mitigate this risk, the agent is designed as a decision-support system, ensuring transparency and maintaining human oversight.

7. Conclusion

This project demonstrates how an intelligent agent can be constructed by integrating supervised learning and reasoning techniques within a sustainability-focused domain. Using the Food Waste dataset, the agent learns patterns, predicts waste trends, and supports informed decision-making. Future work may incorporate reinforcement learning and real-time data streams.